

R0004E Exercise 3

1. Draw a Bode diagram of the open-loop transfer function $G(s)$ of the closed-loop system shown in Figure Q1. Determine the gain margin, phase margin, phase-crossover frequency, and gain crossover frequency with MATLAB.

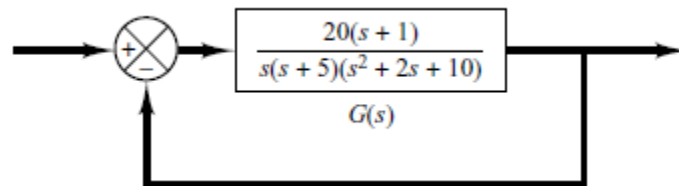


Figure Q1.

2. Given a third-order model of a ship-steering system

$$G_3(s) = \frac{0.05K_a}{s \left(1 + \frac{s}{0.1}\right) \left(1 + \frac{s}{2}\right)}$$

Draw the Bode diagram of the transfer function, for $K_a = 20$. Find the gain margin and the phase margin.

Show that the pole at $s = -2$ can be ignored and the model can be reduced to a second-order transfer function

$$G_3(s) = \frac{0.05K_a}{s \left(1 + \frac{s}{0.1}\right)}$$

3. Give the transfer function $G(s)$ in a unity feedback system

$$G(s) = \frac{K}{(s^2 + 2s + 2)(s + 2)}$$

Determine the value(s) of the gain K that make the system marginally stable.

Find the phase margin and the gain margin of the transfer function at that value(s) of K .